Exercise3

March 1, 2026

1 Exercise 3

[2]:
```
import numpy as np
import warnings
warnings.filterwarnings('ignore')
```

[3]:
```
from ucimlrepo import fetch_ucirepo
breastcancer = fetch_ucirepo(id=15)
spambase = fetch_ucirepo(id=94)
X = breastcancer.data.features
y = breastcancer.data.targets
```

2 3.1

[4]:
```
X.head()
```

[4]:

	Clump_thickness	Uniformity_of_cell_size	Uniformity_of_cell_shape	Marginal_adhesion	Single_epithelial_cell_size	Bare_nuclei	Bland_chromatin	Normal_nucleoli	Mitoses
0	5	1	1	1	2	1.0	3	1	1
1	5	4	4	5	7	10.0	3	2	1
2	3	1	1	1	2	2.0	3	1	1
3	6	8	8	1	3	4.0	3	7	1
4	4	1	1	3	2	1.0	3	1	1

[5]:
```
X.drop(columns=['Single_epithelial_cell_size'], inplace=True)
X.head()
```

[5]:

	Clump_thickness	Uniformity_of_cell_size	Uniformity_of_cell_shape	\
0	5	1	1	
1	5	4	4	
2	3	1	1	
3	6	8	8	
4	4	1	1	

	Marginal_adhesion	Bare_nuclei	Bland_chromatin	Normal_nucleoli	Mitoses
0	1	1.0	3	1	1
1	5	10.0	3	2	1
2	1	2.0	3	1	1
3	1	4.0	3	7	1
4	3	1.0	3	1	1

Single_epithelial_cell_size column is succesfully droped.

3 3.2

[6]:
```
y.head()
```

[6]:

	Class
0	2
1	2
2	2
3	2
4	2

[7]:
```
y = y.replace({2: -1, 4: 1})
y.head()
```

[7]:

	Class
0	-1
1	-1
2	-1
3	-1
4	-1

[9]:
```
np.unique(y)
```

[9]:
```
array([-1,  1])
```

The 2s and 4s are successfully replaced by -1s and 1s.